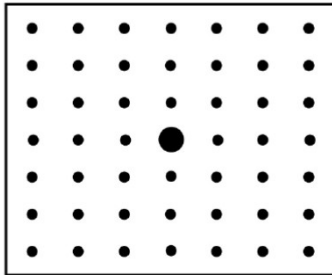


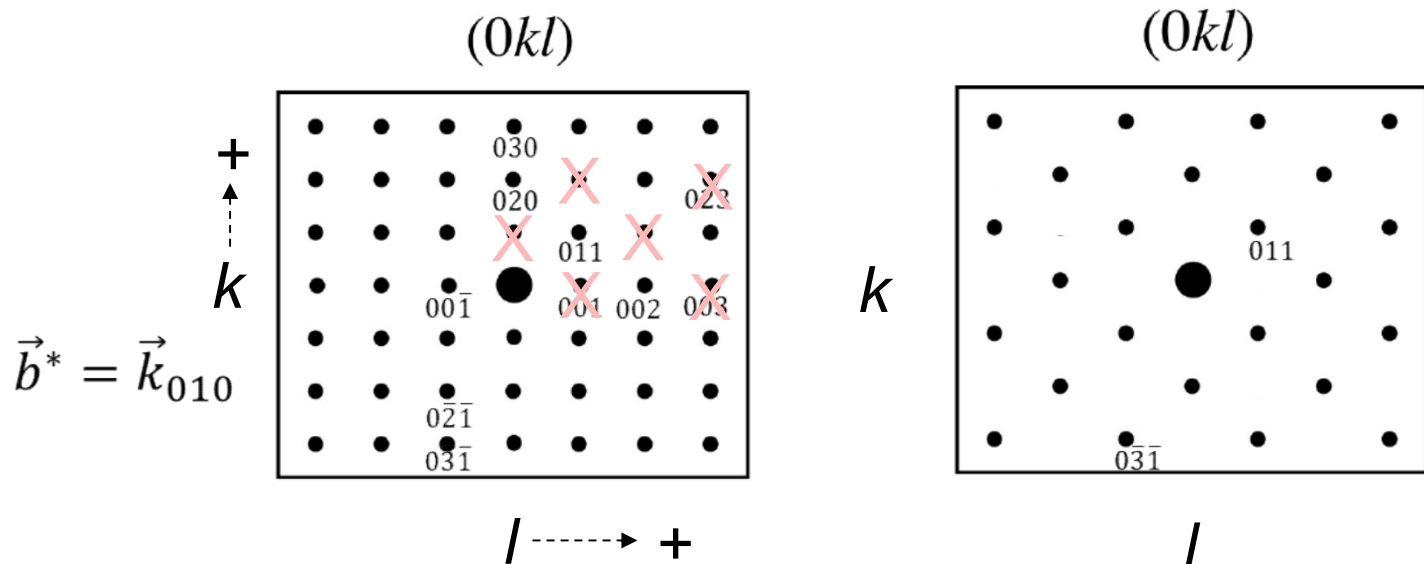
In-class exercise 3 (solution)

Space group: $Pnma$
System: Orthorhombic P



Glide plane	Reflection condition	Reflections involved
a	$h = 2n$	
b	$k = 2n$	$0kl$ for plane $\perp a$
c	$l = 2n$	$h0l$ for plane $\perp b$
n	$h + k, k + l$ or $h + l = 2n$	$hk0$ for plane $\perp c$
d	$h + k, k + l$ or $h + l = 4n$	

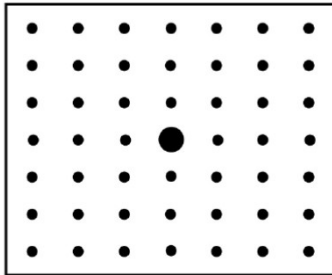
n : diagonal glide $\perp a \rightarrow$ systematic absences in $(0kl)$ reciprocal plane when $k+l=\text{odd}$



In-class exercise 3 (solution)

Space group: $Pnma$

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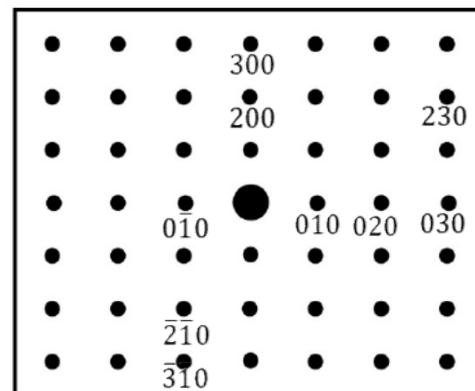
n: diagonal glide $\perp a \rightarrow$ systematic absences in $(0kl)$ reciprocal plane when $k+l=\text{odd}$

m: mirror plane $\perp b \rightarrow$ no systematic absences associated with mirror planes

a: a-glide $\perp c \rightarrow$ systematic absences in $(hk0)$ reciprocal plane when $h=\text{odd}$

a: a-glide $\perp c$

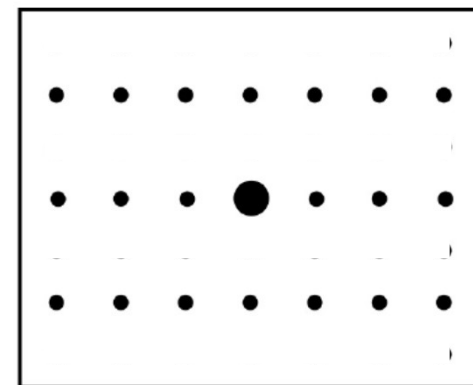
h



$k \rightarrow +$

$(hk0)$

h



k

In-class exercise 3 (solution)

Space group: $Pnma$

System: Orthorhombic P

Glide plane	Reflection condition	Reflections involved
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d	$h + k, k + l$ or $h + l = 4n$	

n: diagonal glide $\perp a \rightarrow$ systematic absences in $(0kl)$ reciprocal plane when $k+l=\text{odd}$

a: a-glide $\perp c \rightarrow$ systematic absences in $(hk0)$ reciprocal plane when $h=\text{odd}$

Check for self-consistency along the three axes $(h00)$, $(0k0)$ and $(00l)$

e.g., n-glide $\rightarrow (0k0)$ and $(00l)$ absences when k and l are odd

e.g., a-glide $\rightarrow (h00)$ absence when h is odd

